

Tyvora Apex: Project Overview

Tyvora Apex is an enterprise-grade Space Situational Awareness (SSA) and orbital risk intelligence platform, conceptually described as a "Bloomberg Terminal for Space." It is designed to provide real-time trajectory visualization, close-approach (conjunction) risk assessment, and an AI-driven advisory system for maneuver synthesis to ensure orbital survival.

Core Capabilities

1. High-Stakes, Low-Latency Rendering

- The platform utilizes a lightweight Presentation Layer to render geospatial environments via a strict-mounted CesiumJS WebGL globe.
- It is heavily optimized via tile culling and GPU conservation techniques to provide instantaneous, stutter-free rendering (60fps) simultaneously with deep predictive physics.

2. SGP4 Propagation & Ephemeris Engine

- The core engine uses `satellite.js` to execute high-frequency Simplified General Perturbations-4 (SGP4) orbital propagation models on live Two-Line Elements (TLE).
- Calculations are transformed from Earth-Centered Inertial (ECI) to Geodetic coordinates.
- Orbital paths are recalculated pseudo-real-time to maintain fidelity and present precise spatial vectors to operators.

3. Autonomous Collision-Resolution & Advisory Unit

- The system separates complex computational physics workloads into an independent Compute Layer backend API, offloading high-resolution collision probability and miss-distance covariance calculations.
- **The Burn-Vector Engine:** When a collision probability breach is detected, the backend executes a Clohessy-Wiltshire (CW) relative motion equation solver. It computes the necessary drift rate and synthesizes the optimal Delta-V vector (in m/s) and burn direction (e.g., PROGRADE) needed for an evasion maneuver.
- **AI Moat (Human-in-the-Loop):** An integrated RAG-based AI "Advisory Unit" ingests raw telemetry, performs contextual historical lookups, and automatically triggers emergency override states when collision probability exceeds 0.01%. It surfaces the physics-backed Delta-V evasion burns to human operators for final "Verify & Execute" authorization.

Architectural Philosophy

- **Cloud-Native SaaS:** By isolating maneuver strategy computation from the spatial rendering loop via an API, Tyvora achieves predictable horizontal scalability for multi-tenant SaaS environments and protects intellectual property algorithms.
- **Mission Control UX:** The interface is stripped of unnecessary web padding. It adopts a high-contrast industrial palette (Charcoal, Teal, Red) and strictly monospaced data grids to present critical information efficiently.

Technical Stack

- **Core Engine:** Next.js 14, React 18, TypeScript
- **3D Visualization:** CesiumJS (WebGL, WebAssembly-accelerated)

- **Physics Modeling:** satellite.js (SGP4/SDP4 mathematical models)
- **UI/UX & Styling:** Tailwind CSS, Framer Motion (Hardware-accelerated Springs)
- **Typography:** Orbitron (Display), JetBrains Mono (Data)
- **Database/Backend:** Supabase (for telemetry data and schema management, as seen in the project root)

In summary, Tyvora Apex represents a paradigm shift in aerospace defense and collision avoidance. It seamlessly combines highly complex orbital physics compute operations, sophisticated AI advisory tools, and top-tier WebGL visualizations into a scalable, high-performance web platform.